



SMOKING
INSIGHT

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Why Do People Smoke?

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Presented to
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A report analyzing psychological constructs
of smoking

INTRODUCTION

01

HISTORY

In 2023, approximately **7 million lives** were lost globally due to smoking, with the U.S. accounting for around **500,000 deaths**. Despite a decline in smoking prevalence from 15-19% during 2011-2015 to roughly 11-12% in 2022, the total number of smokers has risen due to population growth.

Policy initiatives and public health programs have been effective in reducing rates, particularly amongst younger cohorts, while older adults (65+) exhibit a slight increase in smoking trends.



02

AUDIENCE

While the initiatives have been successful the battle is never really over. Since habits like smoking stems from psychological normalization of the act, in this project I wanted to evaluate what constructs of the Health Belief Model (HBM) influence it.

Notably, while smoking rates decline, the absolute number of smokers continues to rise. Insights drawn from this analysis aim to develop effective public policies and messaging surrounding smoking cessation, addressing the psychological factors at play in smoking behavior.

DATA

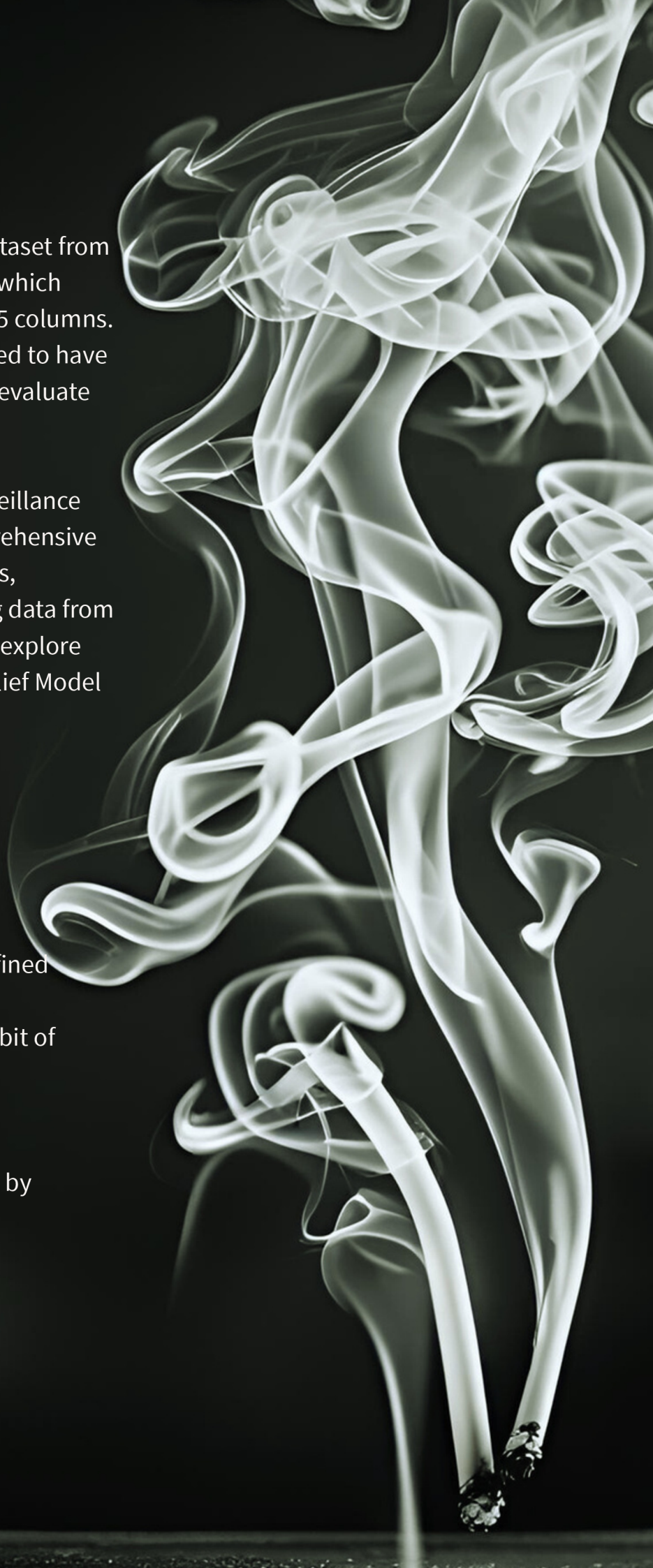
We used Behavioral Risk Factor Surveillance System (BRFSS) Dataset from Kaggle API for years 2011-2015 which contained 2380047 rows and 645 columns. We chose this because we wanted to have as many features as possible to evaluate the predictors of smoking.

The Behavioral Risk Factor Surveillance System (BRFSS) provides comprehensive data on health-related behaviors, including smoking. By analyzing data from 2011-2015, this project seeks to explore the **constructs** of the Health Belief Model influencing smoking habits.

AIM

Question 1: What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? Are there any policy implications?

Question 2: Does this trend vary by age group ?



METHODOLOGY

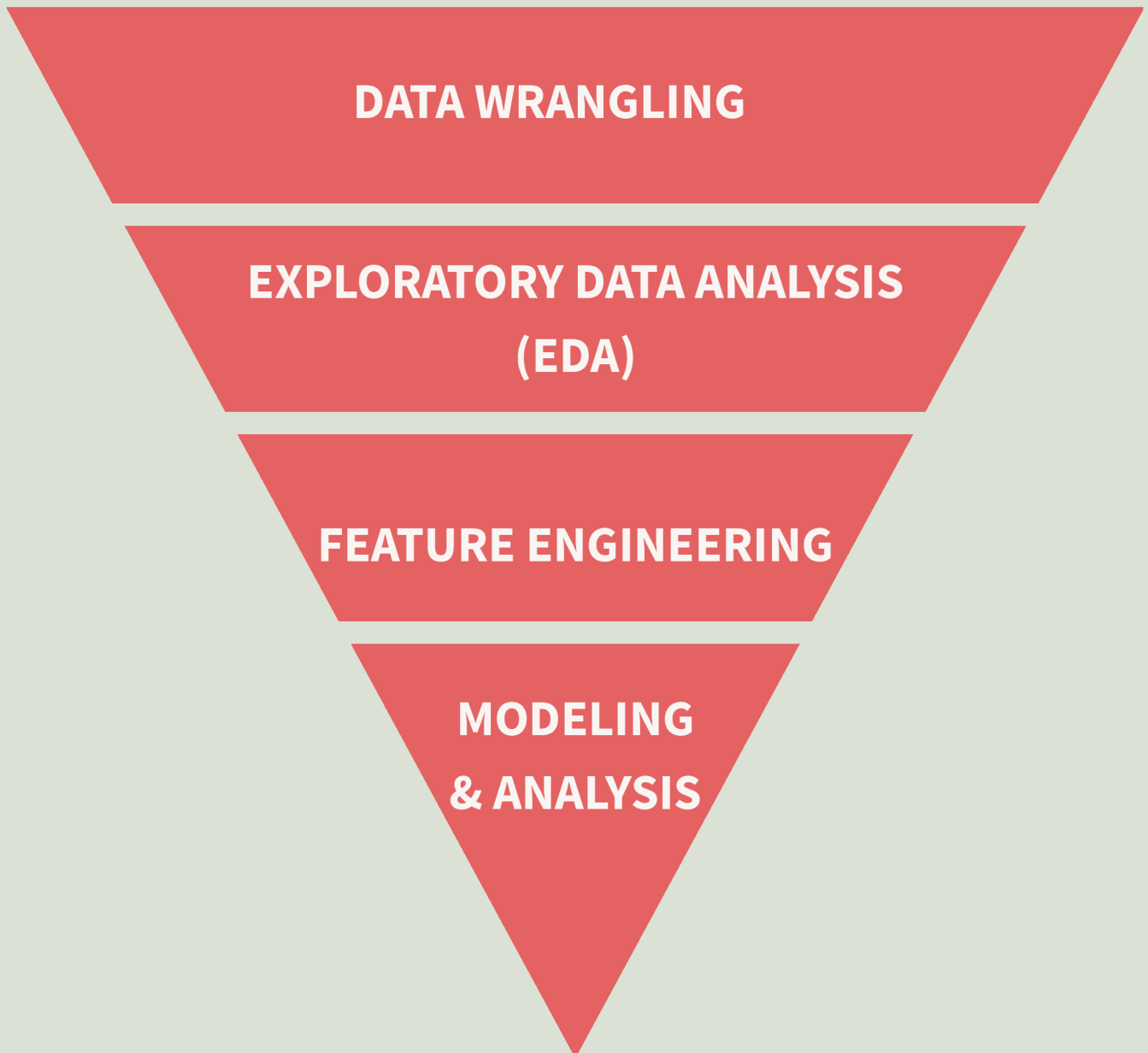
01

Health Belief Model Constructs

The Health Belief Model is a psychological framework developed by Public Health investigators between 1950s-1960s with an attempt to explain and predict health behaviours by focusing on the attitudes and beliefs of individuals.



02 Data Science Process



I chose to build a prediction model with higher Recall and F1-score to be aggressive in identifying smokers. As a next step, these features were mapped to the constructs

DATA WRANGLING

01

The dataset was gathered from telephonic survey so either due to personal reservations of revealing information or due to some issues like lack of time during data collection there were large instances of missing data.

Features with missing values > 0.75 threshold was dropped, this brought feature set from 645 columns to 87 columns

02

As it was a survey dataset, many features served as identifiers of the records and provided less information on smoking habits. These included features like phone number, sequence number.

These features were dropped from the dataset to avoid less interference during data exploration and feature engineering

03

There were many features that captured same information. These included have features like weight in pounds, weight in kgs, calculated weight variable which essential information.

We dropped some features by comparing missing values within each type and retaining the most information feature

04

A lot of features were of type categorical but were represented as float since the values were encoded as codes

This was handled by appropriately converting the features to categorical or ordinal or int/float appropriately.

05

Some features had erroneous values. This included features like year that had some values of 2016 when our analysis was conducted for data collected between 2011-2015.

These issues were corrected by imputing the columns with correct values based on conditional filtering.

06

Some features had values belonging to different metric systems. This included features like height and weight where some record values were represented in inches and some in meters, similarly in kilograms and pounds.

This was handled by conditional value conversion based on the feature type and conversion formulas.

EDA

01

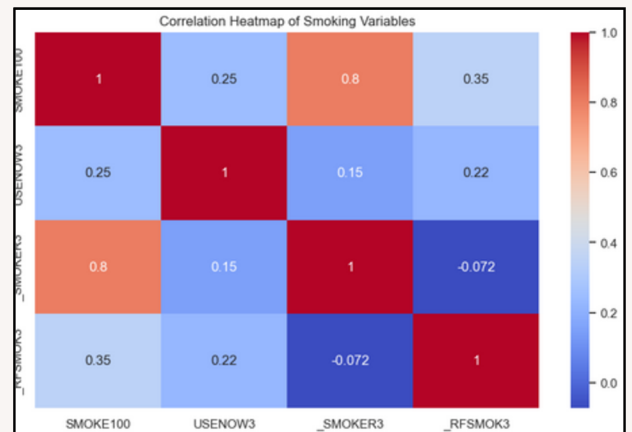
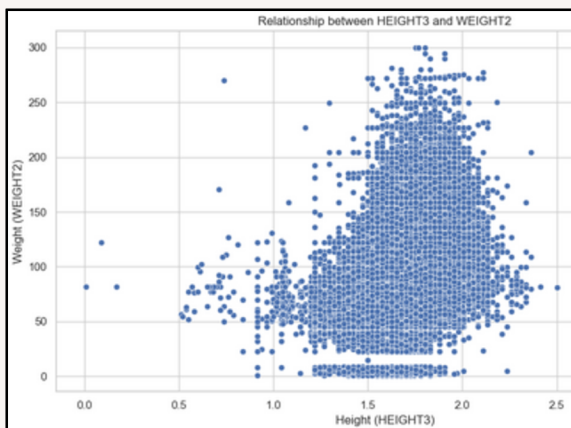
Selection of Target Variable:

Our original aim is to evaluate which features contribute most to identifying a smoker. As surveys usually involve enquiring about a question in multiple forms, there were various labels related to smoking including:

02

Data Validation:

The wrangled dataset has numerical features including height, weight. Since these are demographic features we need to ensure that these values are within acceptable ranges. To ensure data validity we:



03

Map dataset to HBM constructs:

Since we want to identify the psychological and behavioral factors behind smoking, we had to carefully examine the interpretation of features and place them under appropriate HBM construct definitions.

04

Binning zero-imbalance features into fewer buckets:

Variables like number of days of poor mental and physical health have a lot of zeros and to make processing simpler we binned them into manageable buckets.

05

Evaluating Data Leakage:

Variables like number of days of poor mental and physical health have a lot of zeros and to make processing simpler we binned them into manageable buckets.

> Chi-square tests

> Mutual Information

FEATURE ENGINEERING

We one-hot encode categorical and standardize numerical variables as a preparatory step before modeling. This is where we also split our data into training and test sets to be used in the modelling step

MODELING

I wanted to evaluate smoking prediction using 2 algorithms [CatBoost and LightGBM]- both belonging to Gradient Boost algorithms. I opted for gradient boosting since our dataset has a mix of categorical and numeric variables and having an ensemble learning method will yield a stronger predictor by combining many weak learners. Each algorithm is trained with different hyperparameters i.e a base model with default parameters and a tuned model by providing a list of hyperparameters. The two modeling algorithms applied on BRFSS dataset are:

1. CatBoost: I chose to train the original dataset (Features not encoded) using this algorithm since it natively handles categorical values. The training time for this model was on the higher end averaging ~xxxx for both tuned and untuned models. Here's a summary of performance of Base and Tuned Catboost models
2. LightGBM: I used the encoded dataset since this algorithm does better when encoded explicitly before training. This is mainly a choice to see if I will get similar results of Catboost in less time and it successfully met my assumption. Here's a breakdown of model evaluation metrics of Base and Tuned LightGBM models

We select CatBoost Model as our winner because it performed best against other models for Recall and second best in PR-AUC, ROC-AUC, F1-Score compared to other models. It is considerably high in Training time, but is still the second best from the above table

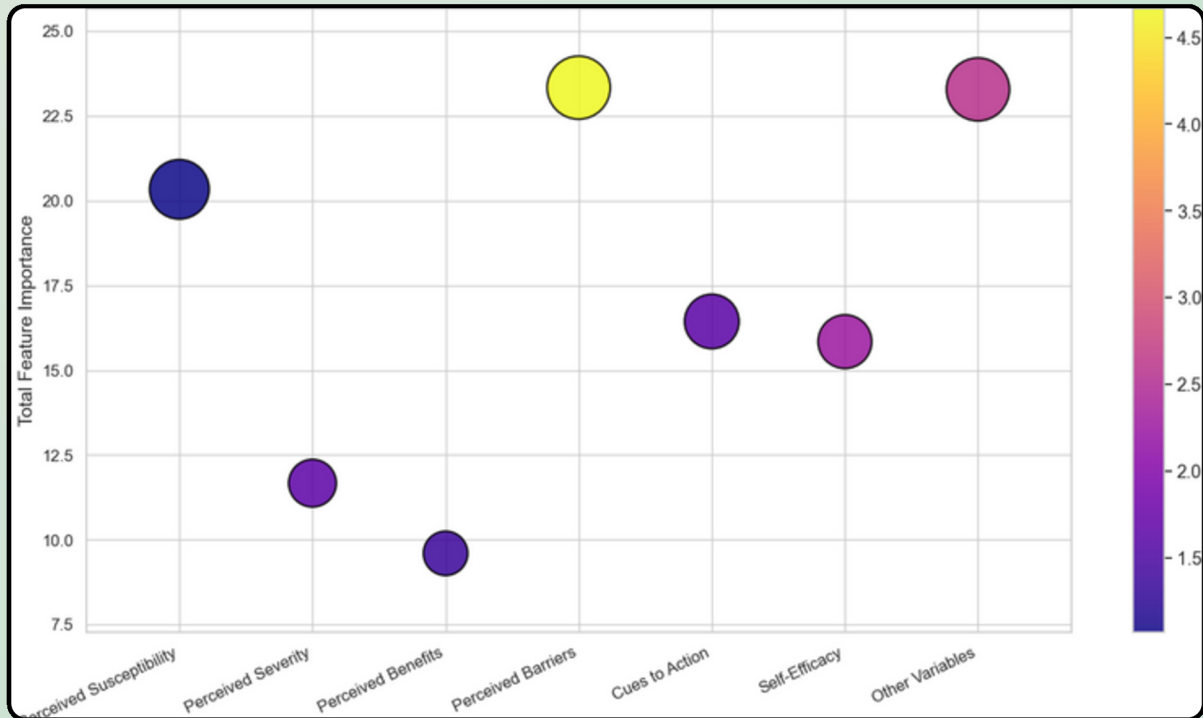
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC	Training Time
CatBoost	0.729721	0.333598	0.713685	0.454670	0.796675	0.458032	1144.751788
CatBoost Tuned	0.730858	0.334517	0.712342	0.455249	0.796907	0.458349	1831.023640
LightGBM	0.728925	0.332754	0.713301	0.453807	0.795863	0.456280	16.839819
LightGBM Tuned	0.730335	0.333869	0.711520	0.454481	0.796365	0.457688	3298.030455

ANALYSIS

01

What constructs defined by Health Belief Model (HBM) contribute most towards the habit of smoking ? What are some policy implications ?

We aggregate total_importance and average_importance scores from top feature set of CatBoost model and plot a bubble plot showing the effect of each construct on the target variable.



1. Which Health Belief Model (HBM) Constructs Mattered Most?

Construct	Total Importance	Avg. Importance	Takeaway
Perceived Barriers	Highest	Highest	Most critical in smoking prediction—barriers like income, education, and cost matter.
Other Variables	Very High	Moderate	Demographics (e.g., age, marital status) still play a large role.
Perceived Susceptibility	High	Low	Many small-risk features (e.g., chronic conditions) together matter.
Cues to Action	Moderate	Moderate	Behavioral triggers like binge drinking and checkups matter.
Self-Efficacy	Moderate	Moderate	Features like physical activity, weight, and BMI help reflect confidence to quit.
Perceived Severity	Lower	Lower	Severity alone (e.g., general health, mental health) had less direct predictive power.
Perceived Benefits	Lowest	Lowest	Although important theoretically, these were not strong predictors in the data.

2. Top Features Driving Smoking Behavior

Rank	Feature	Importance	HBM Construct(s)	Interpretation
1	_EDUCAG	15.5	Perceived Barriers	Lower education limits access to cessation resources.
2	_AGE65YR	9.6	Other	Seniors (65+) showed distinct smoking behavior trends.
3	MARITAL	7.2	Other	Social isolation or support plays a role.
4	_RFBING5	6	Cues to Action	Binge drinking co-occurs with smoking.
5	WEIGHT2	5.1	Self-Efficacy	Indicates belief in personal control over health.
6	CHCCOPD1	4.9	Susceptibility	Respiratory illness increases smoking risk awareness.
7	GENHLTH	4.8	Susceptibility	Self-perceived poor health is a relevant indicator.
8	_INCOMG	4.2	Perceived Barriers	Lower income directly hinders quitting efforts.
9	HEIGHT3	3.5	Self-Efficacy	General health proxy; associated with confidence.
10	_RFBMI5	3.2	Self-Efficacy	Weight management is linked to behavior change confidence.

3. Policy Implications

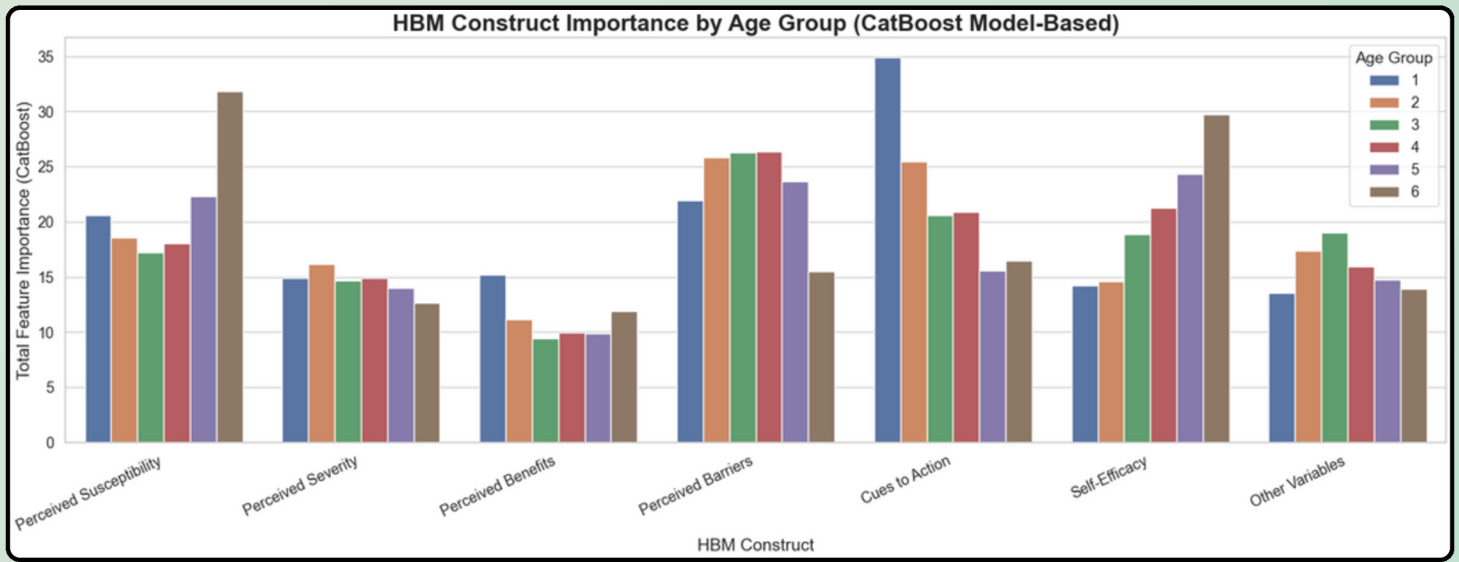
- Perceived Barriers are the most actionable levers for intervention (e.g., cost, education, insurance).
- Demographics (e.g., age, marital status) remain foundational and cannot be ignored.
- Cues to Action like binge drinking or routine checkups signal opportune moments for targeted messaging.
- Self-efficacy indicators (weight, activity) matter more than perceived severity.

This insight strengthens the policy implications of our analysis. Effective anti-smoking strategies should prioritize removing barriers, targeting at-risk groups, and leveraging behavioral triggers—not just warnings about long-term severity.

02

Does this trend vary by age group ?

This grouped bar chart displays how the predictive importance of HBM constructs (as defined in the Health Belief Model) varies by age group. Each construct's total feature importance was calculated using a CatBoost model trained separately on each age group.



1. Age Group Breakdown

Code	Age Range	Description
1	18–24	Young Adults
2	25–34	Early Career Adults
3	35–44	Mid-Career Adults
4	45–54	Mature Adults
5	55–64	Pre-Retirees
6	65+	Seniors

2. Key Observations by Construct

Perceived Susceptibility

- Most predictive in Age 5 (55–64): Chronic conditions raise smoking risk awareness.
- Also high in Age 1 (18–24): Reflects public health education and early awareness.

Perceived Benefits

- Peaks in Age 1: Young people are optimistic about quitting benefits.
- Drops in midlife (3–5): Suggests resignation or competing life priorities.

Perceived Barriers

- Dominates Ages 2–4: Middle-aged groups face practical quitting obstacles (time, stress, cost).
- Declines after 65: Possibly due to access to Medicare or fewer work constraints.

Cues to Action

- Strongest in 18–24 group: Behaviors like binge drinking or doctor visits trigger action.
- Declines steadily with age: Older adults may rely less on external prompts.

Self-Efficacy

- Rises with age, peaking at 65+: Older adults feel more confident in managing health.
- Implies readiness to act if empowered and supported.

Other Variables

- Slightly stronger for mid-aged adults (35–44).
- Includes demographic factors (e.g., marital status, income, education) not captured in core HBM.

3. Policy & Communication Implications

Target Group	Key Strategy
18–24	Most influenced by Cues to Action and Perceived Benefits Leverage peer influence, social media, benefits messaging.
25–54	Dominated by Barriers. Address barriers: stress, time, cost, support access.
55–64	Influenced most by Susceptibility. Emphasize risk and susceptibility via health condition links.
65+	High on Self-Efficacy, Susceptibility, and Benefits. Empower with tools and reinforce confidence to quit.

FUTURE IMPROVEMENTS

- It would love to spend more time creating a dashboard with different tabs explaining the purpose of project, to sample data from the smoking subset and evaluate how we can introduce changes to current policy that will target the right constructs to reduce smoking among people.
- I would also like to add more recent data to my project (~2021-2024) to evaluate differences in smoking habits and differences in constructs over a decade.